PS03 · Liquid & Overnight fund net flows: forecast and cash buffer

Monthly AMFI net flows, 2019-04–2026-09; 24-release rolling backtest (2024-10–2026-09); live origin 2026-09 (data as of 2026-10-13).

- **November 2026 buffer (q75, reading (a))**: Liquid ₹14,697 cr · Overnight ₹21,627 cr
- If the 3× penalty is on the rupee amount (reading (b)), use q95: Liquid ₹64,892 cr · Overnight ₹29,972 cr
- Forecast-based q75 (sarimax_cal) beats the 'last year same month' baseline under both penalty readings in both categories.

Local, reproducible build: `python -m liqfc run` → outputs/*.csv → this deck. All numbers are pulled from those CSVs.

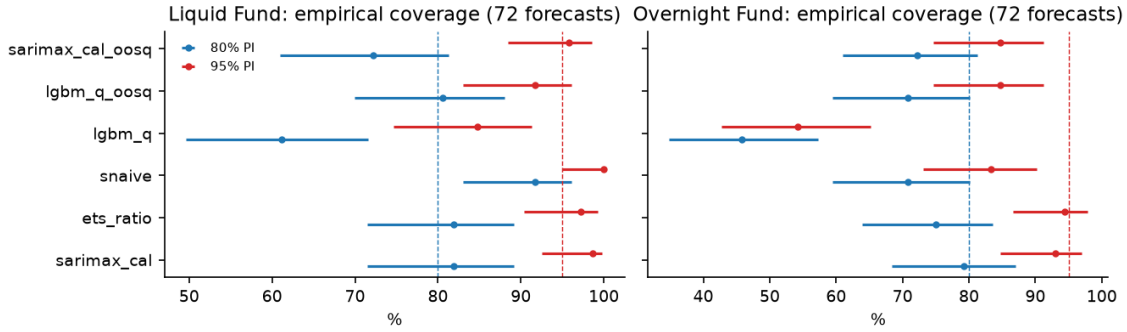
Data and setup

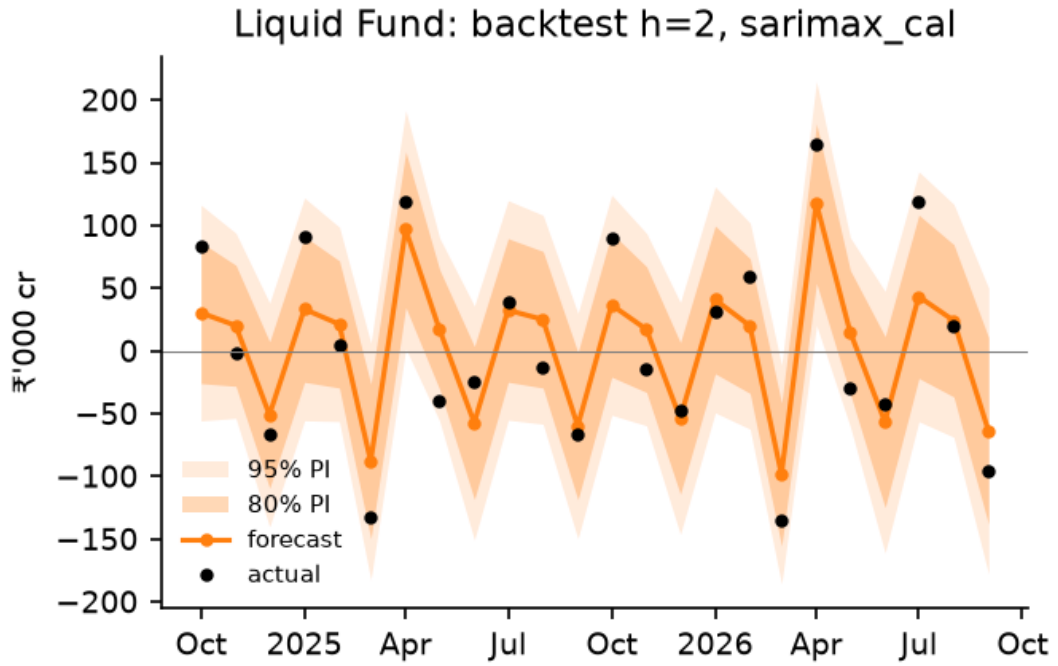
- **AMFI monthly reports** (category sales, redemptions, net flow, AUM), parsed from official files with sha256 manifest; identity sales − redemption = net checked.
- **Drivers:** RBI WACR and net LAF (Money Market Operations), FBIL 3M T-bill, NSE Nifty 50, plus a calendar (advance-tax quarter-ends, FY-end, April).
- **Leakage control:** each origin uses only data released before its as-of date (AMFI release ≈ 9–11th of next month).
- **Decision timing:** the buffer for month M is set on day 1 from the release for M−2, so the decision uses **h = 2**.
- **Models:** seasonal naive, ratio-to-AUM naive, ETS on ratio, SARIMAX with calendar, LightGBM quantile; conformal (out-of-sample quantile) variants. Nested selection: 12 months to pick, 12 to evaluate.

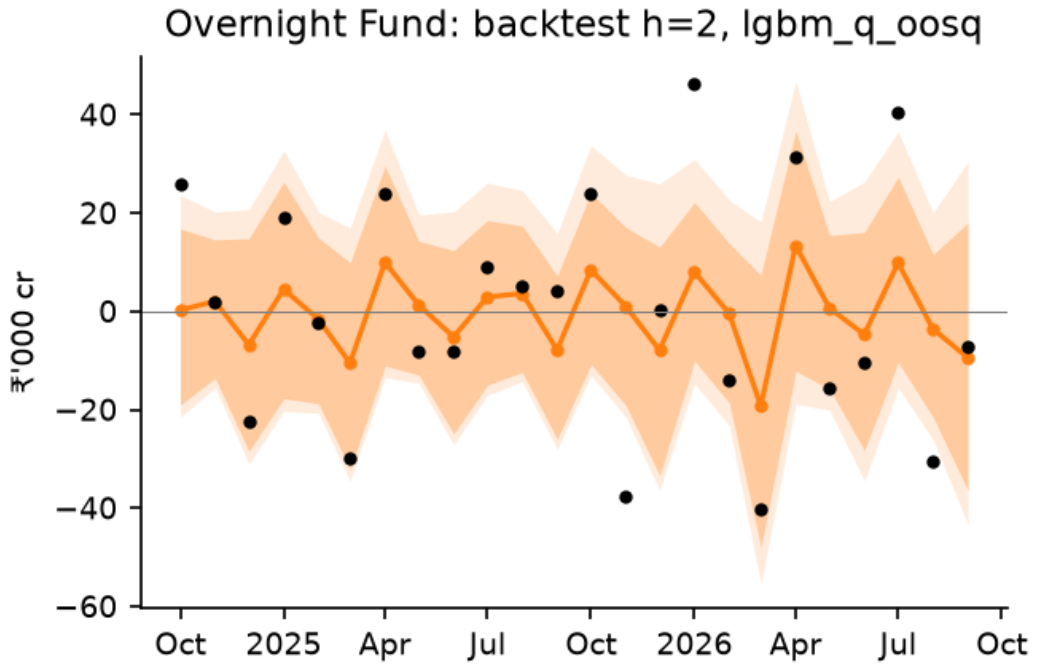
Accuracy and calibration (24 targets × h = 1–3)

Cat.	Model	MASE	Pinball ₹cr	Cov80	Cov95
Liquid	snaive	0.83	7,267	92%	100%
Liquid	sarimax_cal	0.78	6,563	82%	99%
Overnight	snaive	1.15	3,426	71%	83%
Overnight	sarimax_cal	1.02	3,172	79%	93%
Overnight	lgbm_q_oosq	1.08	3,423	71%	85%

Nested selection picks **sarimax_cal** (Liquid) and **lgbm_q_oosq** (Overnight). Seasonal patterns dominate; no model is far ahead of seasonal naive on point accuracy.







Decision cost, penalty reading (a) [primary]

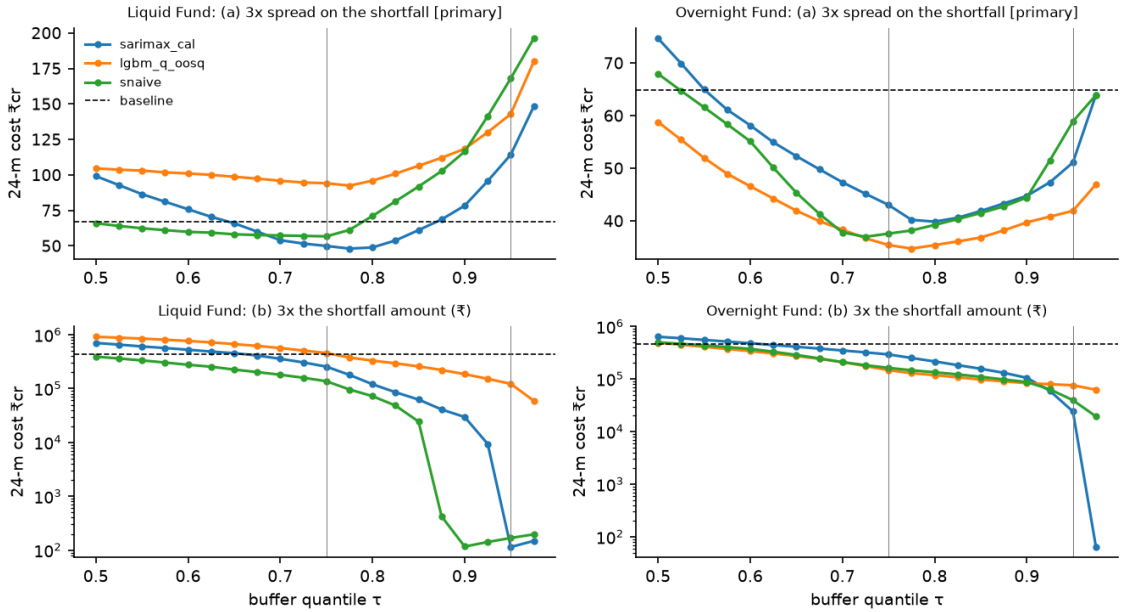
Cost = idle × s/1200 + 3 × shortfall × s/1200, s = FBIL 3M T-bill − WACR (floor 0.10%). h = 2, 24 months, ₹ crore.

Cat.	Policy	24-m cost	Breach months	vs baseline (90% bootstrap CI)
Liquid	baseline	67.1	8	—
Liquid	sarimax_cal q75	50.0	6	₹17.1 cr cheaper than baseline over 24 m (90% CI includes 0)
Liquid	sarimax_cal q95	114.3	0	₹47.2 cr costlier than baseline over 24 m (90% CI excludes 0)
Overnight	baseline	64.9	10	—
Overnight	sarimax_cal q75	43.1	7	₹21.8 cr cheaper than baseline over 24 m (90% CI excludes 0)
Overnight	sarimax_cal q95	51.1	3	₹13.8 cr cheaper than baseline over 24 m (90% CI includes 0)

Under (a) the cost-minimising quantile is 3/(3+1) = 0.75. The spec's q95 holds too much idle cash for Liquid.

The 3× penalty is ambiguous: both readings

Cat.	Model	(a) q75	(a) q95	(b) q75	(b) q95	Flips
Liquid	ets_ratio	66.4	124.0	289,238	19,604	yes
Liquid	lgbm_q	79.0	134.4	536,914	149,613	yes
Liquid	lgbm_q_oosq	94.2	142.9	456,274	122,700	yes
Liquid	sarimax_cal	50.0	114.3	255,233	114	yes
Liquid	sarimax_cal_oosq	53.3	85.3	284,587	38,668	yes
Liquid	snaive	56.7	168.0	136,712	168	yes
Overnight	ets_ratio	42.1	54.6	279,762	40,461	yes
Overnight	lgbm_q	50.9	48.9	386,947	231,981	no
Overnight	lgbm_q_oosq	35.4	41.9	146,666	75,850	yes
Overnight	sarimax_cal	43.1	51.1	292,451	24,119	yes
Overnight	sarimax_cal_oosq	40.8	47.6	171,635	56,810	yes
Overnight	snaive	37.6	58.9	162,508	39,496	yes

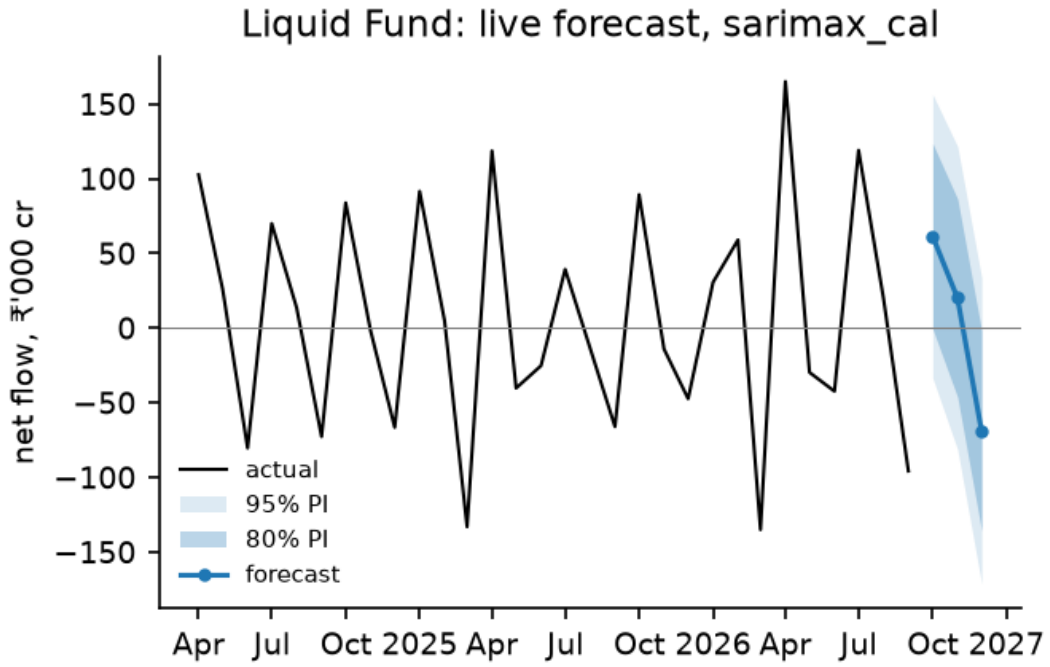


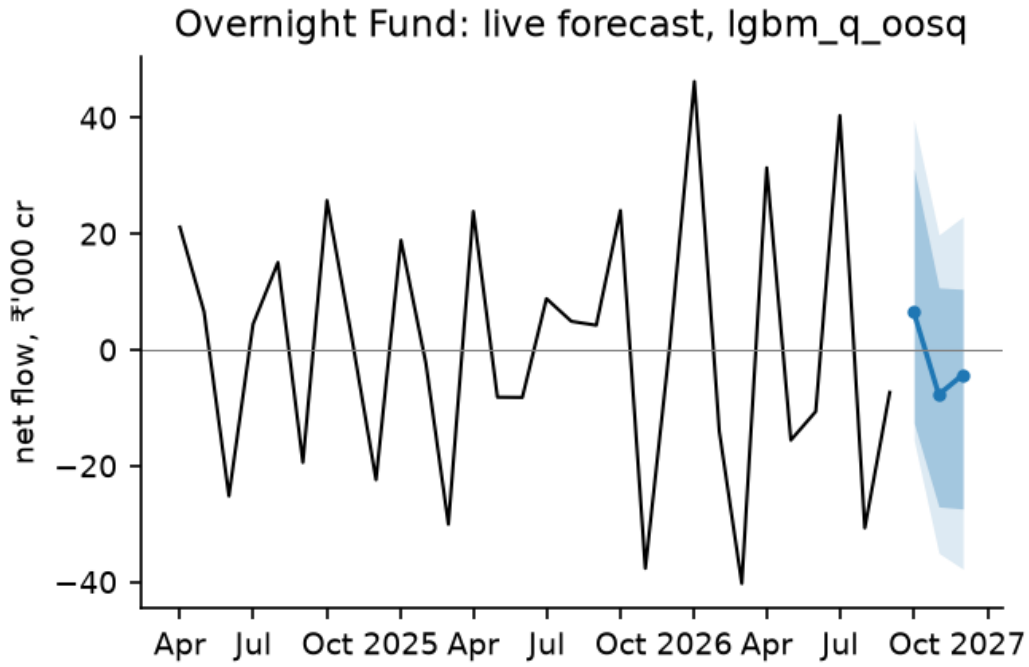
****Yes, the conclusion flips.** Under reading (a) q75 is cheaper than q95 in 11 of 12 category × model cases and q95 in 1; under reading (b) q75 is cheaper than q95 in 0 of 12 category × model cases and q95 in 12. The cheaper policy changes in 11 of 12 cases.**

(a) = 3 × shortfall × spread (primary: same unit as idle cost, critical ratio 0.75). (b) = 3 × shortfall in ₹: about 36,000× the monthly idle carry, so the optimum is ≈ the 99.997th percentile.

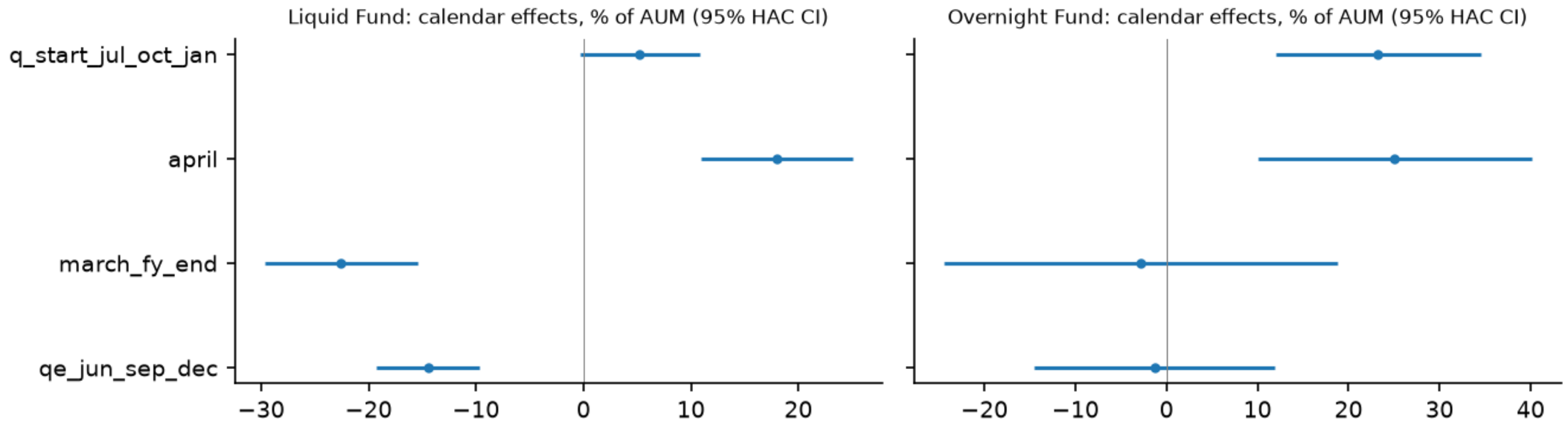
Live forecast from origin 2026-09

Cat.	Model	Month	h	Point	80% PI	Buffer q75	Buffer q95	Baseline
Liquid	sarimax_cal	2026-10	1	61,681	[-361, 123,722]	0	17,948	0
Liquid	sarimax_cal	2026-11	2	20,193	[-46,099, 86,486]	14,697	64,892	14,051
Liquid	sarimax_cal	2026-12	3	-68,770	[-135,644, -1,897]	103,966	154,601	47,308
Overnight	lgbm_q_oosq	2026-10	1	6,510	[-12,496, 31,306]	5,110	13,037	0
Overnight	lgbm_q_oosq	2026-11	2	-7,668	[-26,991, 10,767]	21,627	29,972	37,625
Overnight	lgbm_q_oosq	2026-12	3	-4,250	[-27,322, 10,526]	17,450	32,132	0





What drives the flows



- Calendar effects dominate: advance-tax quarter-ends and March FY-end bring large Liquid outflows; April brings inflows.
- Rates, liquidity and equity variables add no out-of-sample accuracy over calendar + lags (LGBM ablation).
- Overnight: WACR and net LAF are significant contemporaneously, but this is not exploitable at h = 2.

Limits, risks, next steps

- Category-level net flow, not a single fund: scale by the fund's AUM share and its own investor mix.
- 24 test months: bootstrap CIs are wide; Liquid q75 vs baseline under (a) is not conclusive.
- The penalty reading changes the q75 vs q95 choice; confirm the intended definition with the fund.
- The 80% bands under-cover for some models (e.g. conformal SARIMAX Liquid 72%).
- Next: daily or weekly flow data, fund-level holdings, intra-month timing of tax dates.

Full report: reports/report.pdf · Interactive app: python -m liqfc app (localhost:8501).